

MTN South Sudan Mobile Wallet Platform

Bank Escrow Integration Specification

Design Specification





© Ericsson AB 2022

All rights reserved. The information in this document is the property of Ericsson. Except as specifically authorized in writing by Ericsson, the receiver of this document shall keep the information contained herein confidential and shall protect the same in whole or in part from disclosure and dissemination to third parties. Disclosure and disseminations to the receiver's employees shall only be made on a strict need to know basis.

About Ericsson

Ericsson enables communications service providers to capture the full value of connectivity. The company's portfolio spans Networks, Digital Services, Managed Services, and Emerging Business and is designed to help customers go digital, increase efficiency and find new revenue streams. Ericsson's investments in innovation have delivered the benefits of telephony and mobile broadband to billions of people around the world. The Ericsson stock is listed on Nasdaq Stockholm and on Nasdaq New York. www.ericsson.com

Contributors

Name	Role
Abdulhadi Elhussain	Business Analyst

Revision History

Revision	Date	Comments
PA0	09-09-2022	Initial Draft
PA1	16-09-2022	MTN input added
PA2	19-09-2022	Internal Review input added
PA3	01-12-2022	Updated sequence diagrams
PA4	2023-02-21	Added connectivity table
PA5	2023-03-10	Text clean-up



Contents

1	Background information	4
1.1	Introduction	4
1.2	Intended Audience.....	4
1.3	Scope	4
1.4	Assumptions and Constraints	5
1.5	Security.....	5
2	Implementation flow	6
2.1	APIs Integration Flow	6
2.1.1	Use Cases	7
2.1.2	Sample request and Responses.....	10
2.1.3	Error response xml API.....	13
2.2	Regulatory Compliance flow	14
2.2.1	Prerequisites	14
2.2.2	Pre-Conditions.....	14
2.3	Mobile Money Integration into MTN.....	16
2.3.1	Connectivity	16
2.3.2	Client authentication.....	18
2.3.3	Protocol Structure	20
2.3.4	Message Integrity Protection	21
2.3.5	SSL Certificate setup.....	24
3	Annex I.....	25
3.1	API parameters.....	25
3.1.1	transfertoanybankaccount v1.4	25
3.1.2	paymentinstructionresponse v1.0	28
3.1.3	deposit v1.0	29
3.1.4	gettransactionstatus v1.3.....	31
3.1.5	getcustodyaccounttransactionhistory v1.4	32
3.1.6	getaccountholderinfo v2.1	34
3.1.7	getsettlementdepositdetails v2.0	35



1 Background information

With online integration, the settlement information is sent one by one to and from a bank on the XML API. This is sent when a user initiates a withdrawal or a deposit. This allows the user to get immediate response if the transaction is successful, pending, or has failed. The system always sends the settlement information on the XML API. However, the bank has the possibility to send information either on the XML API or as files. To make sure that the settlement information is not processed multiple times, the duplicate detection disallows processing of already-processed settlements. If the communication fails, the Mobile Money Manager retries until an answer is received.

1.1 Introduction

This solution document is for a bank that wants to preform escrow integration into MTN. Escrow integration comprises of two sets of use cases.

- Money deposits and withdrawals.
- Reconciliations and compliance.

MTNs expectation is that deposits are exclusively to Agents (if the profile contains the word agent from the getaccountholderinfo). The mobile money platform identifies users using an FRI (Financial resource Information) in this case all MSISDNS must be sent as FRI:229772121243/MSISDN, **International format is a MUST**.

1.2 Intended Audience

The following are included in the target group for this document:

- System Integrator
- System Administrator
- Application Designer
- Financial Controller

1.3 Scope

- The scope of this Solution is to implement Real-Time XML based integration between Bank and EWP Escrow account.
- Bank will be sending XML request for all deposit requests.
- The bank request will be processed in Real-time, and response provided in Real-time.
- A trust Account will be created and Configured in MTN Mobile Money.



1.4 Assumptions and Constraints

- The bank should implement duplicate detection mechanism for all requests. It will be useful when EWP retries the request to the bank when no response received in the first call
- This integration will be based on asynchronous flows, but it should be able to handle synchronous flows as well. More details in sequence diagrams
- The Bank will open a custody account for MTN and share the details
- MTN South Sudan uses a single BIN approach with its customers implying that all customers in the mobile money platform don't belong to any bank and deposits and withdrawals can be affected from any escrow partner.

1.5 Security

- Bank and EWP will exchange certificates for secure connection.
- The Bank is connecting directly to EWP
- The bank IP will be whitelisted on EWP firewall to execute the following API's
 - deposit v1.0
 - getaccountholderinfo v2.1
 - getcustodyaccounttransactionhistory v1.4
 - getsigningcertificate v1.0
 - gettransactionstatus v1.2
 - paymentinstruction v2.0
 - paymentinstructionresponse v1.0
- No other IP is allowed to send these APIs to EWP unless configured so in the Firewall.
- Two layers of authentication will be used. Presentation and Transport layers.
- Message Signing will be enabled
 - **Presentation:** Presentation layer authentication will be through SSL tunnel. This will be achieved by exchanging certificates.
 - **Transport:** Transport Authentication will be used when the bank connects to the mobile money platform. The bank will hold only both the private key and public key but send the private key in the request.
- The bank will be able to query the status of an account holder as well as the name.



2 Implementation flow

2.1 APIs Integration Flow

- The bank will consume the following EWP APIs for this integration.

API Name	Version	Use Case
deposit	v1.0	Deposit money to Agent account in EWP
getaccountholderinfo	v2.1	Fetch customer details and check if MSISDN presented by Agent is registered
getcustodyaccounttransactionhistory	v1.4	Returns a list of transactions history for that custody account
gettransactionstatus	v1.2	To get status of previously initiated transaction. NOTE that either referenceid OR financialtransactionid must be set.
paymentinstruction	v2.0	Sending a payment instruction request to a bank for withdrawal
paymentinstructionresponse	v1.0	Sending asynchronous response to a payment instruction. This request is sent from the bank to complete a pending payment instruction



2.1.1 Use Cases

2.1.1.1 Withdrawal

This section describes how withdrawals is processed

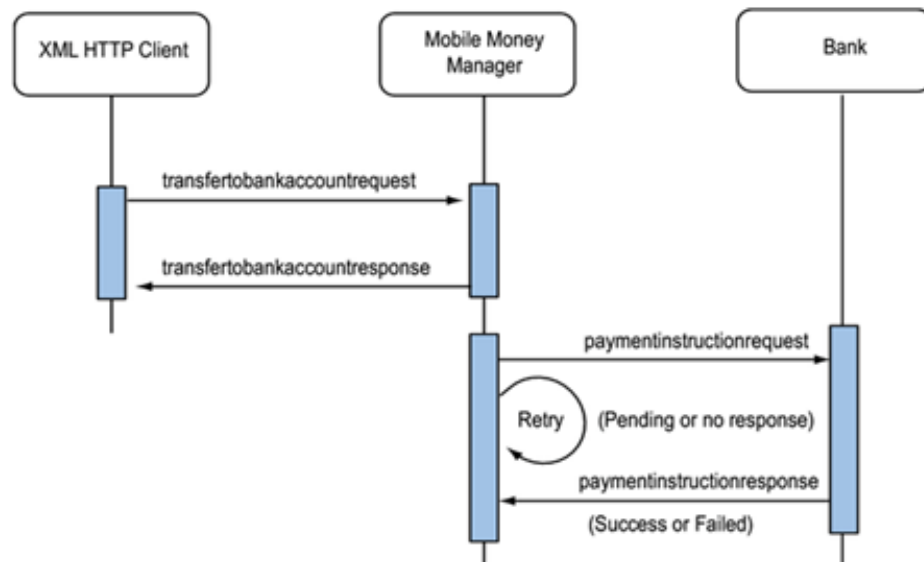


Figure 1 Withdrawal Sequence - Online

1. The user initiates a withdrawal from the system with a transfer to bank account operation (transfertoanybankaccount or transfertobankaccount, refer to XML over HTTP).

Note: this step is initiated by a user (agent or service provider) through EWP USSD menu or EWP partner GUI internal and hence the bank doesn't need to develop this API.

2. The system locates the correct custody account by using the rules defined by the administrator and identifies that the custody account has online settlement configured.
3. The system initiates a paymentinstruction request (including payment instruction details) and sends it to the bank.
4. The bank will process the request and responds with a Success or Failure.
5. If no paymentinstruction response is received or a pending response is received, the Mobile Money Manager continues to retry the same request until a success or failure response received.

Note: Due to the retry mechanism the bank system should be able to detect duplicates and give the right status and not execute the transaction twice.



2.1.1.2

Deposit

This section describes how deposits from the bank are processed

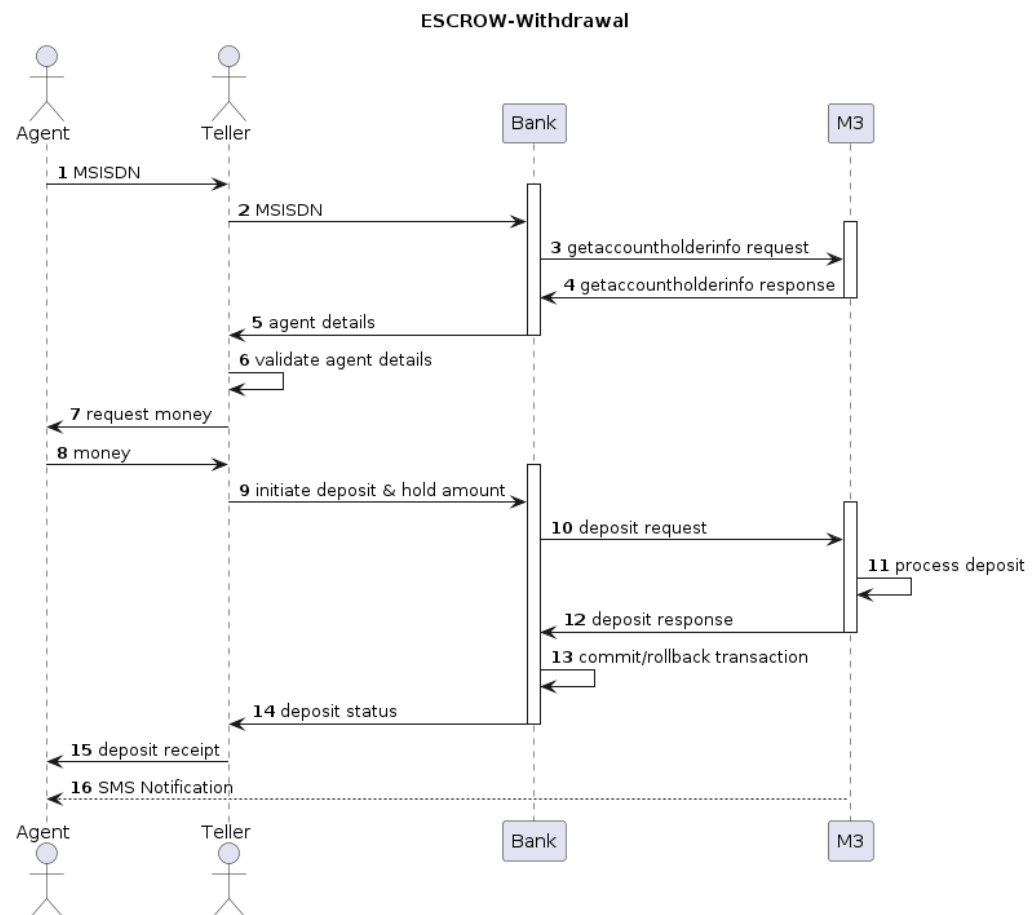


Figure 2 Deposit Sequence Diagram

1. The user (teller) at the bank side inputs Agents MSISDN to validate if it is registered and details of the person in MTN (EWP) before proceeds with deposit. A getaccountholderinfo request is sent from the bank to EWP.
2. EWP will respond with getaccountholderinfo response with the agent details such as name and status.
3. After verifying account details the cash will be collected from the agent and deposited to the escrow account at the bank.
4. Then the bank will initiate deposit request that includes the details in the deposit request.
 - It is possible to add a message and the first name and surname of the receiver since this information is helpful if the receiving account is not found.
5. The Mobile Money Manager responds with a status, SUCCESSFUL, FAILED, or SUSPENDED.



- Normal account rules apply for the deposit processing which means that the deposit fails or be suspended if the user, for example exceeds account limit.
 - Transaction gets failed if account is blocked. Error response is sent to bank and bank also should fail transaction.
 - A financial administrator can resolve a suspended deposit onto the correct account using the Customer Care GUI. This implies bank must mark transaction as **successful when status SUSPENDED** is returned because e-money has been created already.
6. Based on the status the bank either commit the deposit to the escrow account or rollback the transaction.
 7. Deposit receipt will be generated by the bank and shared with the agent
 8. The agent will receive the regular EWP SMS notification with the deposit details.

2.1.1.3

Get Transaction Status

This section describes the sequence for checking status of any transaction performed or committed

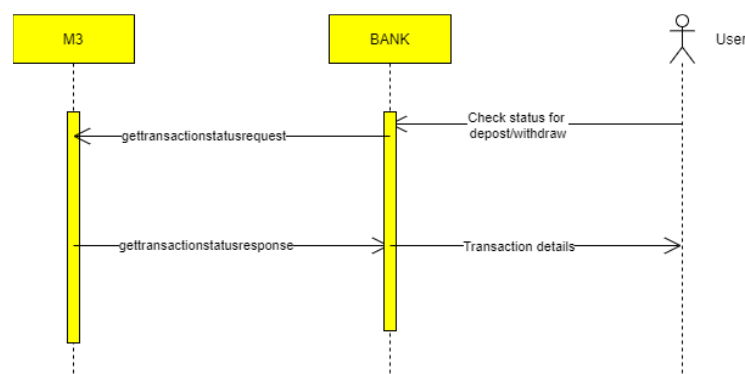


Figure 3 Get Transaction Status Sequence



2.1.1.4 Getcustodyaccounttransactionhistory

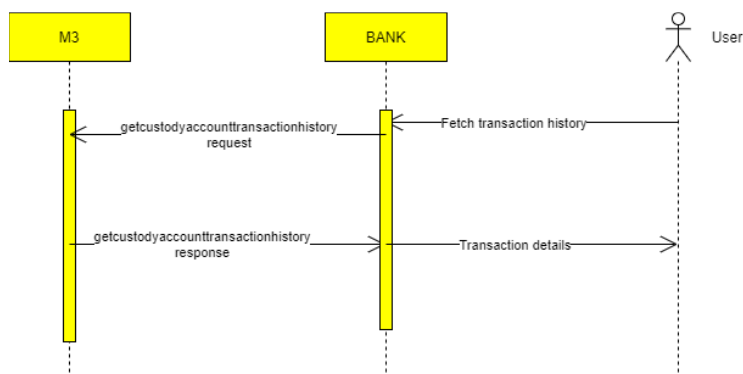


Figure 4 Getcustodyaccounttransactionhistory Sequence

2.1.2 Sample request and Responses

2.1.2.1 deposit API:

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:depositrequest
xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v1_0"> <bankcode>123456</bankcode>
<accountnumber>12345678</accountnumber> <transactiontimestamp> <timestamp>2015-04-
10T00:00:00.000+03:00</timestamp> </transactiontimestamp> <amount> <amount>2000</amount>
<currency>UGX</currency> </amount> <receiver>FRI:256772712580/MSISDN</receiver>
<banktransactionid>3512172222</banktransactionid> <receiverfirstname>Brian</receiverfirstname>
<receiversurname>Ssebagala</receiversurname> <message>My Float</message>
</ns2:depositrequest>
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?> <ns4:depositresponse
xmlns:ns4="http://www.ericsson.com/em/emm/settlement/v1_0"
xmlns:op="http://www.ericsson.com/em/emm/v1_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema"> <status>SUCCESS</status>
<financialtransactionid>226923</financialtransactionid>
<failurereason>failurereason</failurereason> <!-- optional --> </ns4:depositresponse>
```

2.1.2.2 getaccountholderinfo

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:getaccountholderinforequest
xmlns:ns2="http://www.ericsson.com/em/emm/provisioning/v2_1">
<identity>ID:25600000003/MSISDN</identity> </ns2:getaccountholderinforequest>
```

Response:



```
<?xml version="1.0" encoding="UTF-8"?> <ns0:getaccountholderinforesponse
xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v2_1"> <accountholderbasicinfo>
<firstname>NAMtest</firstname> <surname>testing</surname> <profilename>MTNZA LITE KYC
ACCOUNT</profilename> <identityvalues>
<identityvalue>ID:27700000003/MSISDN</identityvalue>
<identityvalue>ID:11340008/ID</identityvalue> </identityvalues> <defaultfri> <defaultfri>
<fri>FRI:8520011/MM</fri> <currency> <code>ZAR</code> </currency> </defaultfri>
</defaultfri> <acceptedtc /> <accountholderstatus>ACTIVE</accountholderstatus>
<bankdomainname>MTNZA</bankdomainname> <hasparent>false</hasparent>
</accountholderbasicinfo> <credentialslist /> <offnet>false</offnet>
</ns0:getaccountholderinforesponse>
```

2.1.2.3 getcustodyaccounttransactionhistory

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:getcustodyaccounttransactionhistoryrequest
xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_3"> <bankcode>1234</bankcode>
<accountnumber>0123456789</accountnumber> <startdatefrom>2019-03-
05T00:00:00Z</startdatefrom> <startdateto>2013-09-05T23:00:00Z</startdateto>
</ns2:getcustodyaccounttransactionhistoryrequest>
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?> <ns6:getcustodyaccounttransactionhistoryresponse
xmlns:ns6="http://www.ericsson.com/em/emm/financial/v1_3"
xmlns:fic="http://www.ericsson.com/em/emm/financial/v1_3/common"
xmlns:ns2="http://www.ericsson.com/em/emm/v1_0/common"
xmlns:op="http://www.ericsson.com/em/emm/v2_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema"> <totalcount>1</totalcount> <historylist>
<entry> <financialtransactionid>2570713</financialtransactionid>
<transactionstatus>SUCCESSFUL</transactionstatus> <transfertype>DEPOSIT</transfertype>
<startdate>2017-04-11T11:44:04.102+03:00</startdate> <commitdate>2017-04-
11T11:44:04.566+03:00</commitdate> <initiatinguser>ID:centbank/ADMIN</initiatinguser>
<realuser>ID:centbank/ADMIN</realuser> <direction>OUT</direction>
<from>FRI:1508144/MM</from> <fromaccount>FRI:1508144/MM</fromaccount> <fromamount>
<amount>2000.00</amount> <currency>UGX</currency> </fromamount> <originalamount>
<amount>2000.00</amount> <currency>UGX</currency> </originalamount>
<to>FRI:256779999508/MSISDN</to> <toaccount>FRI:1498181/MM</toaccount> <toamount>
<amount>2000.00</amount> <currency>UGX</currency> </toamount> <tomessage>My
Float</tomessage> <maininstructionid>10100</maininstructionid>
<instructionid>10023</instructionid> <externaltransactionid>3512172222</externaltransactionid>
<toaccountholder>ID:440728/ID</toaccountholder>
<banktransactionid>3512172222</banktransactionid>
<banktransactiontype>DEPOSIT</banktransactiontype> </entry> </historylist>
</ns6:getcustodyaccounttransactionhistoryresponse>
```

2.1.2.4 getsigningcertificate

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:getsigningcertificaterequest
xmlns:ns2="http://www.ericsson.com/em/emm/messagesigning/v1_0" version="1.0" />
```



Response:

```
<?xml version="1.0" encoding="UTF-8"?> <ns3:getsigningcertificateresponse
xmlns:ns3="http://www.ericsson.com/em/emm/messagesigning/v1_0"
xmlns:msi="http://www.ericsson.com/em/emm/messagesigning/v1_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema"> <signingcertificate>-----BEGIN
CERTIFICATE----- -----END CERTIFICATE-----</signingcertificate> <cachain />
</ns3:getsigningcertificateresponse>
```

2.1.2.5 gettransactionstatus

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:gettransactionstatusrequest
xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_2">
<referenceid>referenceid</referenceid>
<financialtransactionid>financialtransactionid</financialtransactionid>
<identity>identity</identity> <admininitiated>false</admininitiated>
</ns2:gettransactionstatusrequest>
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:gettransactionstatusresponse
xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_2">
<financialtransactionid>financialtransactionid</financialtransactionid>
<status>SUCCESSFUL</status>
<providertransactionid>providertransactionid</providertransactionid>
</ns2:gettransactionstatusresponse>
```

2.1.2.6 paymentinstruction

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:paymentinstructionrequest
xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v2_0"> <transactiontimestamp>
<timestamp>2001-12-31T12:00:00</timestamp> </transactiontimestamp> <amount>
<amount>50.0</amount> <currency>ZK</currency> </amount>
<paymentinstructionid>12345</paymentinstructionid>
<receiverbankcode>00131</receiverbankcode>
<receiveraccountnumber>1231414141441</receiveraccountnumber>
<receiverfirstname>John</receiverfirstname> <receiversurname>Doe</receiversurname>
<message>message</message> </ns2:paymentinstructionrequest>
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:paymentinstructionresponse
xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v2_0"> <status>SUCCESS</status>
<paymentinstructionid>12345</paymentinstructionid>
<banktransactionid>12345</banktransactionid> <transactiontimestamp> <timestamp>2001-12-
31T12:00:00</timestamp> </transactiontimestamp> <bookingtimestamp> <timestamp>2001-12-
31T12:00:00</timestamp> </bookingtimestamp> <amount> <amount>500.0</amount>
<currency>UGX</currency> </amount> </ns2:paymentinstructionresponse>
```



2.1.2.7 paymentinstructionresponse

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:paymentinstructionresponserequest  
xmlns:ns2="http://www.ericsson.com/em/emm/settlement/v1_0"> <status>SUCCESS</status>  
<paymentinstructionid>paymentinstructionid</paymentinstructionid>  
<banktransactionid>banktransactionid</banktransactionid> <transactiontimestamp>  
<timestamp>2001-12-31T12:00:00</timestamp> </transactiontimestamp> <bookingtimestamp>  
<timestamp>2001-12-31T12:00:00</timestamp> </bookingtimestamp> <amount>  
<amount>500.0</amount> <currency>ZMW</currency> </amount>  
</ns2:paymentinstructionresponserequest>
```

Response:

```
<?xml version="1.0" encoding="UTF-8"?> <p:paymentinstructionresponsereponse  
xmlns:p="http://www.ericsson.com/em/emm/settlement/v1_0" />
```

2.1.3 Error response xml API

Below is the sample error response XML.

```
<?xml version="1.0" encoding="UTF-8"?> <ns0:errorResponse  
xmlns:ns0="http://www.ericsson.com/lwac" errorcode="ACCOUNTHOLDER_NOT_FOUND">  
<arguments name="id" value="27700000003" /> </ns0:errorResponse>
```

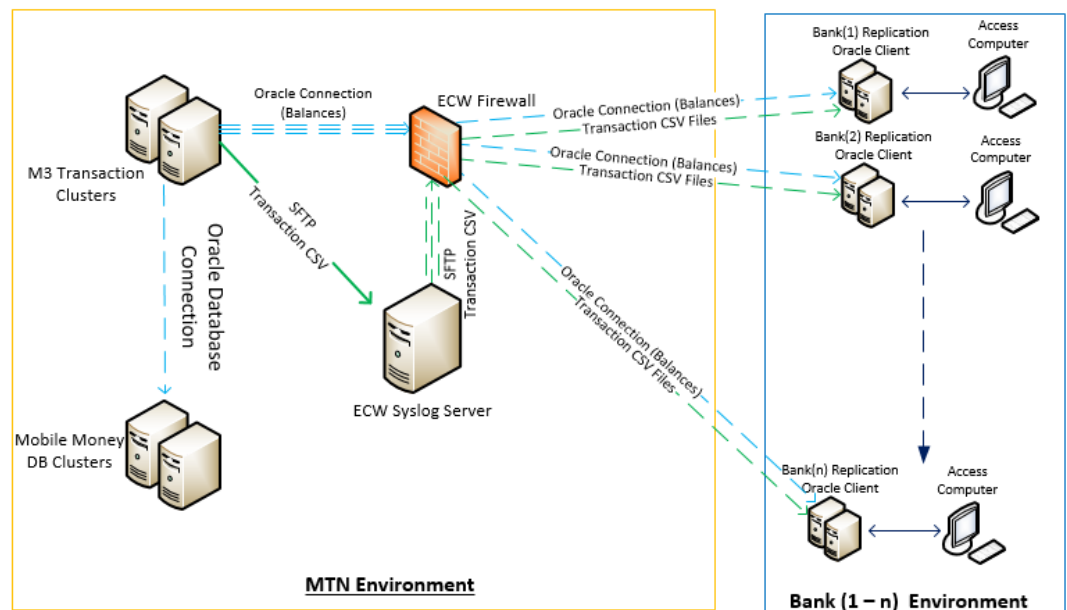
```
<?xml version="1.0" encoding="UTF-8"?> <ns0:errorResponse  
xmlns:ns0="http://www.ericsson.com/lwac" errorcode="COMMUNICATION_ERROR" />
```

```
<?xml version="1.0" encoding="UTF-8"?> <ns2:errorResponse  
xmlns:ns2="http://www.ericsson.com/lwac" errorcode="TRANSACTION_TIMED_OUT" />
```



2.2 Regulatory Compliance flow

From a compliance view, all transactions in the mobile money platform must be replicated to an escrow partner, a server deployed in the bank environment will be updated with balances, but they bank will only have view access to subscriber data from the same. Data is replicated in real time into an oracle environment in the bank.



• Figure 5 Regulatory Compliance Flow

2.2.1 Prerequisites

The bank must be able to handle duplicate transactions and can process transactions on their end with the correct TLS version.

2.2.2 Pre-Conditions

The user should be in activated state.

2.2.2.1 Error response xml API

Below is the sample error response XML.

- `<ns0:errorResponse xmlns:ns0="http://www.ericsson.com/lwac" errorcode="COMMUNICATION_ERROR"></ns0:errorResponse>`
- `<ns2:errorResponse xmlns:ns2="http://www.ericsson.com/lwac" errorcode="TRANSACTION_TIMED_OUT"/>`



- Error codes can vary based on actions by a subscriber or their account or the 3rd party transaction status, hence we have codes like COMMUNICATION_ERROR or VALIDATION_ERROR

2.2.2.2 Error Handling

When an operation results in an operation error, the HTTP connection will respond with the code 500. The operation errors may for example be invalid permissions, invalid parameter values or internal errors of the system. If response code 500 is returned, it is possible to unmarshall the errorresponse object from the error stream of the connection.

The actual error codes are subject to change, so they are currently not included in the description of each operation. Generally, the text for an error code allows a person to interpret the error correctly. For example, the code ACCOUNTHOLDER_NOT_FOUND indicates that an identity was provided that does not have a match in the system.

The error code RESOURCE_NOT_FOUND similarly indicates that the requested financial resource, such as an account in the specified currency, does not exist in the system.

Note:

Prevent leaking too much information to an end user of the system. For example, when requesting a voucher that does not exist, the error VOUCHER_NOT_FOUND is returned. When requesting a voucher that exists but with another owner, the error PERMISSION_DENIED is returned. These two errors must not be displayed to an end user since they expose a possibility to use brute force to get hold of valid voucher tokens in the system.

Below is the table for error handling

errorResponse				
Name		Type	Multiplicity	Description
errorcode		xs:string	Mandatory	Refer to future list of error codes.
arguments		tns:errorargumenttype	Optional, 0..N	See errorArguments.
errorArgument				
	name	xs:string	Mandatory	Additional info depending on error code.
	value	xs:string	Mandatory	Additional value depending on error code.



2.3 Mobile Money Integration into MTN

2.3.1 Connectivity

The below URLs will be used for integration into the Mobile Money system, this will be done through a VPN.

Inbound traffic: The URI will depend on the API and is similar to the API request, e.g getbalance request the URI becomes getbalance, debitrequest becomes debit etc.

- Test Platform:

IP:port/bank/deposit

- Production Side:

IP:port/bank/deposit

Outbound traffic: All traffic from the mobile money platform to the 3PP will also have a URI appended at the end of the URL, if a 3PP has a URL like <https://3ppendpoint.com/> and we are sending a payment or getfinancialresource information request to them, the final end point will be <https://3ppendpoint.com/deposit> or <https://3ppendpoint.com/paymentinstruction>



Integration	ESCROW Integration with Equity Bank		
Incoming			
Src IP's			
EXTVIP	10.250.151.39	F5 Port	XXXX
Internal Node	M3	Port	
APIs	Name	Context	Version
	deposit	bank	V1.0
	getaccountholderinfo	bank	V2.1
	gettransactionstatus	bank	V1.3
	getcustodyaccounttransactionhistory	bank	V1.1
	paymentinstructionresponse	bank	V1.0
	getsettlementdepositdetails	bank	V2.0
Allowed Metachar	.,()!?:/+*='_- (space character)		
Certificate	MTN Intermediate CA		
3PP certificate subject to be whielisted on F5	E = CN = OU = O = L = S = C =		
Outgoing			
Internal Node:	M3		
OUTVIP:	10.250.151.38	Port F5:	XXXX
Dst Ip:		Dst Port:	
Certificate	MTN Intermediate CA		



2.3.2 Client authentication

The following authentication methods are supported:

Presentation layer - server-side certificate

Authentication of the Mobile Money Manager service is supported through an **HTTPS server certificate**. In order to successfully connect to the interface, this certificate has to be trusted by the client, either directly, or through a certificate trust chain.

Note: In MTN South Sudan, MTN will be the Certificate Authority who will sign the issued certificates from both Bank & Ericsson side & it will be up to both the Banks & Ericsson to exchange such certificates and deploy them on their firewalls to be trusted.

No matter if server-side certificates are used or not, the Application (the Service Provider system or the Service Bus) still needs to authenticate itself with a **username** and a **password**. There are two methods supported in the Mobile Money Manager for this.

2.3.2.1 Application layer – normal authentication

Authentication of clients of the service is performed by a username and password-based login service, described in 2.3.2.1.1

The system returns the following in the HTTP header for the response:

Set-Cookie: sessionid=default8fcee064690b45faa9f8f6c7e21c5e5a

The session ID is then used as a cookie in the HTTP header in the following requests towards the Mobile Money Manager.

Cookie: sessionid=default8fcee064690b45faa9f8f6c7e21c5e5a

2.3.2.1.1 Login Service

A successful login to the system creates a session ID. The **session ID** can be used for multiple operations and will be kept alive and have its lifetime extended as long as operations are performed within the session lifetime.

Identifying oneself to the system by entering one's credential.

loginrequest:



loginrequest			
Name	Type	Multiplicity	Description
identity	String	Mandatory	Identity of user.
credential	credential information	Mandatory	Credential used to verify the users identity. (refer to section 2.3.2.1.1.1 for details)

- Table 1 Login Request format

loginresponse:

An **empty response**. If HTTP response code 200 is returned the login is successful. If HTTP response code 302 is returned either a secondarylogin with an OTP or a changepassword are expected is needed before successful login. Which one is indicated in the response header. (refer to section 2.3.3.2 for different error codes)

2.3.2.1.1.1 credential information

credential			
Name	Type	Multiplicity	Description
secret	string	Mandatory	Credential secret
type	string	Mandatory	Credential secret format. Must be set to string.

2.3.2.2 Application layer - basic authentication

The protocol also supports basic authentication scheme as defined in RFC 1945. This can be used instead of session establishment for performing an operation without requiring login.

The credentials need to be passed in the HTTP header in the following format:

Authorization: Basic QWxhZGRpbjpvcGVuIHNlc2FtZQ==

Where the credentials are base64 encoded and in the format
<login>:<password>.



2.3.3 Protocol Structure

This section describes the structure of the protocol.

2.3.3.1 URI Syntax

The **URI of the HTTP request** is structured according to the following pattern:

*URI := **/**{context}/{method}*

where

- context - is a fixed string representing the routing entity e.g., /poextvip
- method is a fixed string representing the operation type like debit,payment,sptransfer etc.

The same method can be used by different users. Depending on the user's authentication, a different context is used.

2.3.3.2 Response Status Codes

The protocol uses the regular HTTP status codes/ Status codes in the **200 range** imply that the **request completed normally**. In this case the response body will be formatted according to the expected response syntax for the specified method. Status code in the **300 range** indicate that the **request must be resent to a different location**. Status codes in the **400 or 500 ranges** imply that there was **an error executing the request**.

Used HTTP response codes	
Response Code	Description
200 (OK)	Sent when the request completed successfully.
202 (Accepted)	The request has been accepted for processing, but the processing has not been completed (pending).
302 (Moved Temporarily)	The URL provided in the Location header field indicates where to send the following request and which request type to use. The following request types are possible:
401 (Unauthorized)	The request needs authentication.
403 (Forbidden)	The request was denied and will be denied also in the future.
404 (Not found)	The resource specified in the URI was not found.
500 (Internal Server Error)	The server encountered a general error during execution, see Section 57.

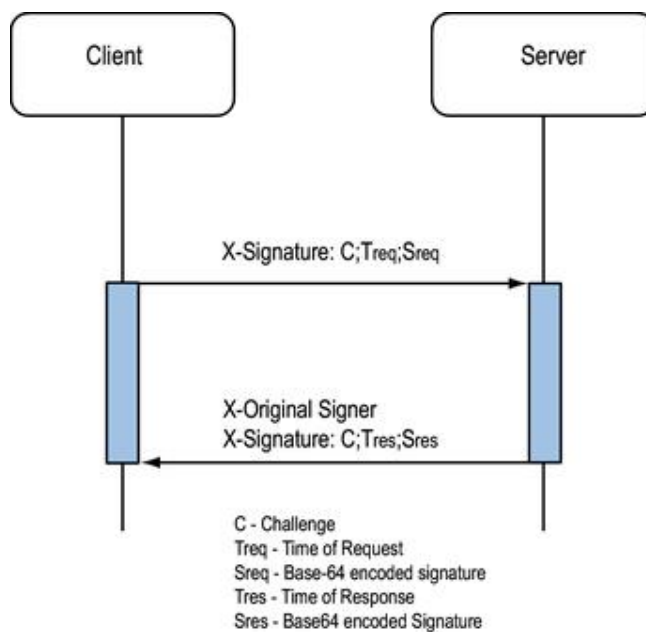
- Table 2 Error Code Table



2.3.4 Message Integrity Protection

This section describes how the message integrity is protected by message signatures.

Message signing is implemented so that a random challenge, a timestamp and selected values from the request or response messages, are included in the signature. The signature string is then delivered together with the message in a HTTP request or response header named X-Signature



• Figure 6 Request and Response Signing

The signature header string consists of the three following parts, separated by semicolon:

- **random-challenge**, a random string chosen by the client, unique during the message life-time. The server will reject messages with the same random-challenge.
- **timestamp**, the number of seconds since 1 Jan 1970, 00:00:00 UTC (also known as POSIX time).
- **b64-encoded-signature**, the Base64 encoded signature.

The signature is created by using the **SHA256 digest** together with an RSA encryption over the UTF-8 encoding of the *to-be-signed* string.

The *to-be-signed* string is composed of the three following parts, separated by a semicolon.



- **random-challenge**, the same random challenge string as in the signature header string.
- **timestamp**, the same time stamp string as for the signature header string.
- **values-to-be-signed**, are made up of zero or more values from the request or response message elements. The values are specific to each message and are listed in each section in this protocol specification.

For cases when no values are to be part of the signature, the to-be-signed string is composed as:

- *random-challenge;timestamp;*

Note:

- The **string** ends with a **semicolon**.
- If message signing is enabled on an incoming request context or outgoing client, all messages must be signed. That also applies for the messages that do not contain any message element values to be included in the signature.

2.3.4.1 Message Signing

This section describes an example flow of the message signing procedure.

Below is an example of a request, and its associated response:

```
<request>
  <transactionid>123456</transactionid>
  <providertransactionid>AREF2014015</providertransactionid>
  <newbalance>
    <amount>12300</amount>
    <currency>EUR</currency>
  </newbalance>
  <paymenttoken>tokenstring1</paymenttoken>
  <message>Thanks for all the fish</message>
  <status>COMPLETED</status>
</request>
```

The client sending the request chooses the data to be used:

- *random-challenge: 2siqin0aj4kbsisqj7cs8s4u5*
- *timestamp-request: 1402997614*

The request requires the following values to be signed, transactionid, providertransactionid, status, the to-be-signed string will look like the following:

2siqin0aj4kbsisqj7cs8s4u5;1402997614;123456;AREF2014015;COMPLETED



This is converted to bytes, using **UTF-8 encoding**, and signed with the clients private key, using **SHA-256 as hash algorithm**. The resulting bytes are then converted to **Base-64** encoding, for example:

b64-encoded-signature:

ZHNmZ25ma2dramdsbHNkZnNrZnllZ2hmamZnaGRnbGdkaGdm

The resulting HTTP request header will look like the following:

X-Signature: 2siqin0aj4kbsisij7cs8s4u5;1402997614;

ZHNmZ25ma2dramdsbHNkZnNrZnllZ2hmamZnaGRnbGdkaGdm

When the server responds to the request, it sends the following response:

<response>
</response>

The response signature is calculated like below:

- random-challenge: 2siqin0aj4kbsisij7cs8s4u5 (same value as in the request)
- timestamp-response: 1402997615 (servers own timestamp)

As this response contains no parameter values to be signed, the to-be-signed string will look like:

2siqin0aj4kbsisij7cs8s4u5;1402997615;

The to-be-signed string is converted to bytes, using UTF-8 encoding, and signed with the servers private key, using SHA-256 as hash algorithm. The resulting bytes are then converted to Base-64 encoding, for example:

- b64-encoded-signature: dzQ1Njd1M2tsajkwNHRlcnR0NDIzMDIIZQ==

The resulting response header looks like:

X-Signature:

2siqin0aj4kbsisij7cs8s4u5;1402997615;dzQ1Njd1M2tsajkwNHRlcnR0NDIzMDIIZQ==

2.3.4.2 Original Signer Header

A system responding to a signed client request from the Mobile Money Manager have to reply with a signed response.

In order to indicate which system, it is, the response must also contain another HTTP header, called X-Original-Signer.



The value of this header is the identity of the system, for example **ID:<bankusername>/USER**. This value is then used by the Mobile Money Manager to look up the corresponding public key for the account holder and verify the response signature.

2.3.5 SSL Certificate setup

Follow the following steps for TLS security setup, this is a guideline is used to ensure we have TLS setup, we use the Mobile Money certificate authority to sign the certificates of the entity integrating to the Mobile Money Platform.

- A 3rd party creates a TLS certificate signing request (CSR) that is to be signed by the MTN Security
- MTN signs that TLS CSR with Mobile Money Certificate Authority (i.e., MTN) and share CSR response certificate as well as CA file with 3pp
- 3pp needs to import that certificate and CA in their keystore as trusted ones
- TLS v1.2 or higher should be used
- On the mobile money platform, mutual authentication will be enabled to ensure only trusted entities are allowed

** Certificate renewal tracking is the responsibility of the 3pp

2.3.5.1 CSR creation guidelines for Mobile Money

- Other security requirements for TLS.
- Signature Algorithm must be SHA256 or above
- Private keys MUST be created and stored within the service host i.e., Importing of Private keys is NOT Allowed.
- Only public keys or certificates can be Exported, Private keys Must not be Exported.
- All certificates MUST be issued for a specific service or host (e.g., ewp.ug.com or mobileapp.ug.com) and SHOULD NOT contain any wild cards.
- All Subject Fields must be fully filled, Mandatory Fields CN,OU, O,L,ST,C and E
- All connection (incoming & outgoing) must terminate at the EWP firewall (F5) to enable traffic inspection.
- Deny all traffic in EWP Firewall by default (Blacklist) and traffic shall be permitted only on a need basis (Whitelist).
- All connections must use TLS1.2 and above
- All 3PP and operator's IT systems must use TLS mutual authentication



- TLS encryption keys must follow minimum requirements for key generation (e.g., key size of at least 2048 bits)

3 Annex I

3.1 API parameters

3.1.1 transfertoanybankaccount v1.4

Transfers money to any bank account. [This operation supports quotation](#),
Classification: **Undefined**.

URI: /{baseContext}/transfertoanybankaccount

Namespace: http://www.ericsson.com/em/emm/financial/v1_4

Parameters

[transfertoanybankaccountrequest](#):

transfertoanybankaccountrequest									
Name	Type	Optionality	Description						
amount	moneydetailstype v1.0 Contains the amount and currency of a monetary value. <table><tr><th>Name</th><th>Type</th></tr><tr><td>amount</td><td>decimal</td></tr><tr><td>currency</td><td>string</td></tr></table>	Name	Type	amount	decimal	currency	string	Mandatory	The amount to transfer. Parameter cannot be NULL. Validated with IsPositiveAmount.
Name	Type								
amount	decimal								
currency	string								
sendingfri	string	Mandatory	This data type represents a Financial Resource Identifier of sending account. Parameter can not be NULL. Validated with IsFRI.						
receivingaccountnumber	string	Mandatory	The receiving bank's account number. Validated with NotBlank. Validated with IsBankAccount.						
receivingbankcode	string	Optional	The receiving bank's bank code. Validated with IsBankCode.						
sendernote	string	Optional	A descriptive note for sender transaction history. Validated with Length (max=256). Validated with IsRestrictedString.						
receivermessage	string	Optional	A descriptive note for receiver bank history. Validated with Length (max=256). Validated with IsRestrictedString.						



receiverfirstname	string	Optional	The first name of the receiver. Validated with IsPersonalName. Validated with Length (max=48).
receiversurname	string	Optional	The surname of the receiver. Validated with IsPersonalName. Validated with Length (max=512).
receivermsisdn	string	Optional	The MSISDN of the receiver. If provided a notification will be sent to the receiver. Validated with IsMSISDN. Validated with Length (max=16).
quoteid	quoteid v1.0	Optional	The ID of the quote to use.
maxnumberofretries	integer	Optional	Number of operations retries. Validated with IsIntegerWithinRange (min=0).

transfertoanybankaccountresponse:

returns the financial transaction id, status, fee, and loyalty.

transfertoanybankaccountresponse												
Name	Type	Optionality	Value	Description								
fee	decimal	Optional		Fee associated with the transaction.								
loyalty	loyalty v1.0	Optional		Loyalty information for the sender.								
financialtransactionid	financialtransactionid v1.0	Mandatory	Contains a financial transaction ID. String	The financial transaction ID associated with the transaction. Parameter cannot be NULL.								
financialtransactionstatus	financialtransactionstatus v1.0	Mandatory	The status of a financial transaction. <table><tr><th>Value</th><th>Description</th></tr><tr><td>FAILED</td><td>Failed transaction.</td></tr><tr><td>PENDING</td><td>Pending transaction.</td></tr><tr><td>SUCCESSFUL</td><td>Successful transaction.</td></tr></table>	Value	Description	FAILED	Failed transaction.	PENDING	Pending transaction.	SUCCESSFUL	Successful transaction.	The status of the financial transaction. If the transaction for some reason fails, an ErrorResponse will be returned, and the HTTP status will be set to 500. Parameter cannot be NULL.
Value	Description											
FAILED	Failed transaction.											
PENDING	Pending transaction.											
SUCCESSFUL	Successful transaction.											

3.1.1.1 paymentinstruction v2.0

Sends a payment instruction request to a bank with details about the transaction that should be performed.

Classification: **Undefined.**

URI: /{baseContext}/paymentinstruction

Namespace: http://www.ericsson.com/em/emm/settlement/v2_0

Schema file: paymentinstruction.xsd



Parameters

paymentinstructionrequest:

paymentinstructionrequest			
Name	Type	Optionality	Description
transactiontimestamp	datetime v1.0	Mandatory	Date and time. Validated with IsDateTimeValue. Parameter can not be NULL.
amount	moneydetailstype v1.0	Mandatory	The amount that should be withdrawn. Parameter cannot be NULL. Validated with IsPositiveAmount.
paymentinstructionid	string	Mandatory	Payment instruction ID. Validated with NotBlank. Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).
receiverbankcode	string	Optional	The receiver's bank code. Validated with IsBankCode.
receiveraccountnumber	string	Mandatory	The receiver's account number. Parameter cannot be NULL. Validated with IsBankAccount.
receiverfirstname	string	Optional	The receiver's first name. Validated with IsFirstname.
receiversurname	string	Optional	The receiver's surname. Validated with IsSurname.
message	string	Optional	Optional message which will appear in the receiver's transaction history. Validated with Length (max=100). Validated with IsRestrictedString.
receivermsisdn	string	Optional	The msisdn of the receiving party. Validated with IsMSISDN.
transmissioncounter	string	Mandatory	A counter representing the number of times a particular request has been transmitted. The counter is set to 1 the first time the request is transmitted and is incremented by 1 for each retransmission of the request. Validated with NotBlank. Validated with IsIntegerStringWithinRange (min=0).
transactionid	string	Mandatory	A unique identifier for the financial transaction. Validated with NotBlank. Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).

paymentinstructionresponse:

paymentinstructionresponse			
Name	Type	Optionality	Description
status	paymentinstructionresult v2.0	Mandatory	Status of the payment instruction from the bank. Parameter can not be NULL.
paymentinstructionid	string	Mandatory	Payment instruction ID as received from EWP. Validated with NotBlank. Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).
banktransactionid	string	Mandatory	The bank's transaction ID. Validated with NotBlank. Validated with Length (max=100). Validated with IsExternalReferenceString.
transactiontimestamp	datetime v1.0	Mandatory	Date and time as sent in the payment instruction request. Parameter can not be NULL. Validated with IsDateTimeValue.
bookingtimestamp	datetime v1.0	Mandatory	Date and time when the bank made the withdrawal. Parameter can not be NULL. Validated with IsDateTimeValue.
amount	moneydetailstype v1.0	Mandatory	The amount that has been withdrawn. Parameter can not be NULL. Validated with IsPositiveAmount.



3.1.2 paymentinstructionresponse v1.0

Sending asynchronous response to a payment instruction. This request is sent from the bank or AM to complete a pending payment instruction.

Classification: **Undefined.**

URI: `/[baseContext]/paymentinstructionresponse`

Namespace: http://www.ericsson.com/em/emm/settlement/v1_0

Parameters

paymentinstructionresponserequest:

paymentinstructionresponserequest												
Name	Type	Optionalit y	Value	Description								
status	paymentinstructionresult v2.0	Mandatory	The status of the payment instruction <table><tr><th>Value</th><th>Description</th></tr><tr><td>FAILED</td><td>The payment instruction failed to be processed</td></tr><tr><td>PENDING</td><td>The payment instruction has not been processed yet</td></tr><tr><td>SUCCESS</td><td>The payment instruction has been successfully processed</td></tr></table>	Value	Description	FAILED	The payment instruction failed to be processed	PENDING	The payment instruction has not been processed yet	SUCCESS	The payment instruction has been successfully processed	Status of the payment instruction from the bank.
Value	Description											
FAILED	The payment instruction failed to be processed											
PENDING	The payment instruction has not been processed yet											
SUCCESS	The payment instruction has been successfully processed											
paymentinstructionid	string	Mandatory		Payment instruction ID as received from EWP. Validated with NotBlank . Validated with Length (max=32). Validated with IsLongString (allowNegativeValue=false).								
banktransactionid	string	Mandatory		The bank's transaction ID. Validated with NotBlank . Validated with Length (max=100). Validated with IsExternalReferenceString .								
transactiontimestamp	datetime v1.0	Mandatory	Contains a time stamp given as date and time. The time represented with an XMLGregorianCalendar. The lexical representation is for example "yyyy-MM-ddTHH:mm:ss.SSS" for millisecond precision, but "yyyy-MM-dd" for date only is also sufficient. Parameter cannot be NULL. <table><tr><th>Name</th><th>Type</th></tr><tr><td>timestamp</td><td>date</td></tr></table>	Name	Type	timestamp	date	Date and time as sent in the payment instruction request. Parameter cannot be NULL. Validated with IsDateTimeValue .				
Name	Type											
timestamp	date											
bookingtimestamp	datetime v1.0	Mandatory		Date and time when the bank made the withdrawal. Parameter cannot be NULL. Validated with IsDateTimeValue .								



amount	moneydetailstype v1.0	Mandatory	Contains the amount and currency of a monetary value. <table><tr><th>Name</th><th>Type</th></tr><tr><td>amount</td><td>decimal</td></tr><tr><td>currency</td><td>string</td></tr></table>	Name	Type	amount	decimal	currency	string	The amount that has been withdrawn. Parameter cannot be NULL. Validated with IsPositiveAmount .
Name	Type									
amount	decimal									
currency	string									

paymentinstructionresponseresponse:

An empty response indicates success.
Values to be signed: None

3.1.3 deposit v1.0

Sent by the bank to the system with detailed information about the deposit that should be performed. This operation supports reservation.

Classification: **Undefined.**

URI: `/{{baseContext}}/deposit`

Namespace: http://www.ericsson.com/em/emm/settlement/v1_0

Parameters

depositrequest:

Table depositrequest			
Name	Type	Optionality	Description
bankcode	bankcode v1.0 String	Optional	Custody account bank code. This along with custody account number may be used to identify the custody account.
accountnumber	accountnumber v1.0 String	Mandatory	Custody account number. This along with custody bank code will be used to identify the custody account. Parameter cannot be NULL.
senderbankcode	bankcode v1.0 String	Optional	Sender account bank code. This along with sender account number may be used to identify the sender's bank account.
senderaccountnumber	accountnumber v1.0	Optional	Sender account number. This along with sender bank code will be used to identify the sender's bank account.
transactiontimestamp	datetime v1.0	Mandatory	Date and time when the transaction was initiated. Parameter cannot be NULL. Validated with IsDateTimeValue .



	<table><tr><th>Name</th><th>Type</th></tr><tr><td>timestamp</td><td>date</td></tr><tr><td></td><td></td></tr></table>	Name	Type	timestamp	date						
Name	Type										
timestamp	date										
amount	moneydetailstype v1.0 <table><tr><th>Name</th><th>Type</th></tr><tr><td>amount</td><td>decimal</td></tr><tr><td>currency</td><td>string</td></tr><tr><td></td><td></td></tr></table>	Name	Type	amount	decimal	currency	string			Mandatory	Amount to be deposited to the receiver. Parameter can not be NULL. Validated with IsPositiveAmount .
Name	Type										
amount	decimal										
currency	string										
receiver	string	Mandatory	A valid FRI referring to the receiver of the deposit. Parameter can not be NULL. Validated with Length (max=100). Validated with IsFRI .								
banktransactionid	string	Mandatory	The bank's transaction ID. Validated with NotBlank . Validated with Length (max=100). Validated with IsExternalReferenceString .								
receiverfirstname	string	Optional	The receiver's first name. Validated with IsFirstname .								
receiversurname	string	Optional	The receiver's surname. Validated with IsSurname .								
message	string	Optional	Optional message which will appear in the receiver's transaction history. Validated with Length (max=100). Validated with IsRestrictedString .								

depositresponse:

depositresponse

Name	Type	Optionality	Description																		
status	depositstatus v2.0 <table><tr><th>Value</th><th>Description</th></tr><tr><td>CREATED</td><td>Deposit created but not yet processed.</td></tr><tr><td>FAILED</td><td>Deposit failed in processing. More details specified in the failure reason attribute. Money not received.</td></tr><tr><td>INACTIVE</td><td>Deposit is about to be removed from the database.</td></tr><tr><td>RESOLVED</td><td>A receiver has been identified, and money moved from suspense account to the receiver's account.</td></tr><tr><td>REVERSED</td><td>A deposit sender has been identified and money moved from suspense account to the sender's account.</td></tr><tr><td>SUCCESS</td><td>Deposit processed successfully.</td></tr><tr><td>SUSPENDED</td><td>Deposit processed, receiver cannot receive money due to financial limits (block status, threshold and so on). Money placed in the suspense account.</td></tr><tr><td>UNRESOLVED</td><td>Deposit processed, but no receiver could be identified. Money placed in the suspense account.</td></tr></table>	Value	Description	CREATED	Deposit created but not yet processed.	FAILED	Deposit failed in processing. More details specified in the failure reason attribute. Money not received.	INACTIVE	Deposit is about to be removed from the database.	RESOLVED	A receiver has been identified, and money moved from suspense account to the receiver's account.	REVERSED	A deposit sender has been identified and money moved from suspense account to the sender's account.	SUCCESS	Deposit processed successfully.	SUSPENDED	Deposit processed, receiver cannot receive money due to financial limits (block status, threshold and so on). Money placed in the suspense account.	UNRESOLVED	Deposit processed, but no receiver could be identified. Money placed in the suspense account.	Mandatory	The status of the deposit.
Value	Description																				
CREATED	Deposit created but not yet processed.																				
FAILED	Deposit failed in processing. More details specified in the failure reason attribute. Money not received.																				
INACTIVE	Deposit is about to be removed from the database.																				
RESOLVED	A receiver has been identified, and money moved from suspense account to the receiver's account.																				
REVERSED	A deposit sender has been identified and money moved from suspense account to the sender's account.																				
SUCCESS	Deposit processed successfully.																				
SUSPENDED	Deposit processed, receiver cannot receive money due to financial limits (block status, threshold and so on). Money placed in the suspense account.																				
UNRESOLVED	Deposit processed, but no receiver could be identified. Money placed in the suspense account.																				
financialtransactionid	string	Optional	Contains the financial transaction ID. Validated with Length (max=40). Validated with IsPositiveLong (allowZero=false).																		
failurereason	string	Optional	Contains the failure reason if status is SUSPENDED. Validated with Length (max=200). Validated with																		



			IsRestrictedString
--	--	--	------------------------------------

3.1.4 gettransactionstatus v1.3

To get status of previously initiated Payment, Transfer or Debit operation. NOTE that either referenceid OR financialtransactionid must be set.

Classification: **Undefined.**

URI: /{baseContext}/gettransactionstatus

Namespace: http://www.ericsson.com/em/emm/financial/v1_3

Parameters

[gettransactionstatusrequest:](#)

gettransactionstatusrequest			
Name	Type	Optionality	Description
referenceid	referenceid v1.0 Contains the External reference id value (string)	Optional	Reference ID which has been entered by user, while initiating the transaction.
financialtransactionid	financialtransactionid v1.0 Contains a financial transaction ID(String)	Optional	Financial ID of the transaction, received in the previously initiated operation.
identity	identity v1.0 Contains an Identity (ID)(String)	Optional	A valid identity associated with the account holder that performed the requested transaction.

[gettransactionstatusresponse:](#)

The transaction status

gettransactionstatusresponse			
Name	Type	Optionality	Description
financialtransactionid	financialtransactionid v1.0 Contains a financial transaction ID(String)	Mandatory	Id for the financial transaction. Parameter cannot be NULL.
status	financialtransactionstatus v1.0	Mandatory	Payment Status. Parameter can not be NULL.



	<table><tr><th>Value</th><th>Description</th></tr><tr><td>FAILED</td><td>Failed transaction.</td></tr><tr><td>PENDING</td><td>Pending transaction.</td></tr><tr><td>SUCCESSFUL</td><td>Successful transaction.</td></tr></table>	Value	Description	FAILED	Failed transaction.	PENDING	Pending transaction.	SUCCESSFUL	Successful transaction.		
Value	Description										
FAILED	Failed transaction.										
PENDING	Pending transaction.										
SUCCESSFUL	Successful transaction.										
providertransactionid	providertransactionid v1.0 Contains the provider transaction id value(String)	Optional	Id for the financial transaction at the provider								
instrumentprovidertransactionid	externaltransactionid v1.0 Contains the External transaction id value(String)	Optional	Transaction id of the external instrument provider								
substatus	transactionsstatus	Optional	Substatus of transaction								

3.1.5 getcustodyaccounttransactionhistory v1.4

Returns a list of transactions matching the parameters in the request.

Classification: **Undefined.**

URI: /{baseContext}/getcustodyaccounttransactionhistory

Namespace: http://www.ericsson.com/em/emm/financial/v1_4

Parameters

getcustodyaccounttransactionhistoryrequest:

getcustodyaccounttransactionhistoryrequest									
Name	Type	Optionality	Description						
custodyaccountidentifier	accountidentifier v1.0 Information to identify an external account used for withdraw and deposit. <table><tr><th>Name</th><th>Type</th></tr><tr><td>accountnumber</td><td>string</td></tr><tr><td>clearingnumber</td><td>string</td></tr></table>	Name	Type	accountnumber	string	clearingnumber	string	Mandatory	The custody account for which to fetch history. Parameter cannot be NULL.
Name	Type								
accountnumber	string								
clearingnumber	string								
paginginfo	paginginfo v1.0 Contains the maximum number of items (default 50) to return in a response, together with the offset at which to start returning items (default 0, i.e. the first item). <table><tr><th>Name</th><th>Type</th></tr><tr><td>maximumnumberofitems</td><td>maximumnumberofitems v1.0</td></tr><tr><td>indexoffset</td><td>indexoffset v1.0</td></tr></table>	Name	Type	maximumnumberofitems	maximumnumberofitems v1.0	indexoffset	indexoffset v1.0	Optional	The maximum number of reservations to retrieve and index to start retrieving from. If unspecified, default paging info will be used.
Name	Type								
maximumnumberofitems	maximumnumberofitems v1.0								
indexoffset	indexoffset v1.0								
financialtransactionid	long	Optional	Retrieve transaction history with this financial transaction ID. Validated						



			with IsPositiveLong (allowZero=false).								
externaltransactionid	externaltransactionid v1.0 Contains the External transaction id value(String)	Optional	Retrieve transaction history with this external transaction ID.								
transactionstatus	financialtransactionstatus v1.0 <table><tr><th>Value</th><th>Description</th></tr><tr><td>FAILED</td><td>Failed transaction.</td></tr><tr><td>PENDING</td><td>Pending transaction.</td></tr><tr><td>SUCCESSFUL</td><td>Successful transaction.</td></tr></table>	Value	Description	FAILED	Failed transaction.	PENDING	Pending transaction.	SUCCESSFUL	Successful transaction.	Optional	Retrieve transaction history with this status.
Value	Description										
FAILED	Failed transaction.										
PENDING	Pending transaction.										
SUCCESSFUL	Successful transaction.										
financialinstructiontypes	list< financialinstructiontype v2.1> Shows the financial transaction type(String)	Mandatory	Retrieve transaction history with this transaction type. It is possible to have more than one transactiontype element to filter on. Shows the financial transaction type. Validated with NotNullElements .								
transactioncreatedinterval	datetimeinterval v1.0 Contains a start date time and an end date time, together making up an interval. <table><tr><th>Name</th><th>Type</th></tr><tr><td>startdatetime</td><td>iso8601datetime v1.0</td></tr><tr><td>enddatetime</td><td>iso8601datetime v1.0</td></tr></table>	Name	Type	startdatetime	iso8601datetime v1.0	enddatetime	iso8601datetime v1.0	Optional	Retrieve transaction history that was created between this interval.		
Name	Type										
startdatetime	iso8601datetime v1.0										
enddatetime	iso8601datetime v1.0										
transactionfinalizedinterval	datetimeinterval v1.0 <table><tr><th>Name</th><th>Type</th></tr><tr><td>startdatetime</td><td>iso8601datetime v1.0</td></tr><tr><td>enddatetime</td><td>iso8601datetime v1.0</td></tr></table>	Name	Type	startdatetime	iso8601datetime v1.0	enddatetime	iso8601datetime v1.0	Optional	Retrieve transaction history that was finalized (SUCCESSFUL, FAILED) between this interval.		
Name	Type										
startdatetime	iso8601datetime v1.0										
enddatetime	iso8601datetime v1.0										
direction	direction v1.0 Used to select transactions to (IN) or from (OUT) an FRI. <table><tr><th>Value</th><th>Description</th></tr><tr><td>IN</td><td>Transactions to an FRI.</td></tr><tr><td>OUT</td><td>Transactions from an FRI.</td></tr></table>	Value	Description	IN	Transactions to an FRI.	OUT	Transactions from an FRI.	Optional	Retrieve transaction history with the specified direction, that is, incoming or outgoing transfers.		
Value	Description										
IN	Transactions to an FRI.										
OUT	Transactions from an FRI.										
otherfri	frivalue v1.0 Contains a Financial Resource Identifier (FRI).(String)	Optional	The FRI of the other party in transfer, could be sender or receiver depending on direction.								
quoteid	quoteid v1.0	Optional	The ID of the quote to use.								

getcustodyaccounttransactionhistoryresponse:

getcustodyaccounttransactionhistoryresponse



Name	Type	Optionality	Description
totalcount	integer	Optional	Total amount of custody account transaction history entries available with the specified condition.
entries	list<getcustodyaccounttransactionhistoryentrytype v1.4>	Mandatory	Contains a list of transactions, each transaction contains type, amount, date and so on. Contains the main summary of a custody account transaction history event. Validated with NotNullElements .
loyalty	loyalty v1.0	Optional	Loyalty information generated with the transaction.

3.1.6 getaccountholderinfo v2.1

Returns some basic information on the account holder connected to the identity. The difference from getaccountholder is that it is not as much information.

Classification: **Undefined**.

URI: `/[baseContext]/getaccountholderinfo`

Namespace: http://www.ericsson.com/em/emm/provisioning/v2_1

Parameters

getaccountholderinfo request:

getaccountholderinfo request			
Name	Type	Optionality	Description
identity	identity v1.0 <i>Contains an Identity (ID). (String)</i>	Optional	A valid identity associated with the account holder. If omitted, the logged in user details are returned.
includeoffnet	boolean	Optional	True indicates that external users not registered in the Mobile Money Manager shall be included in the search. Default value is false. Default value: false.

getaccountholderinfo response:

getaccountholderinfo response			
Name	Type	Optionality	Description
accountholderbasicinfo	accountholderbasicinfotype v1.3	Optional	Contains a subset of account holder information.



	<table><tr><th>Name</th><th>Type</th></tr><tr><td>firstname</td><td>string</td></tr><tr><td>surname</td><td>string</td></tr><tr><td>profilename</td><td>string</td></tr><tr><td>employeeid</td><td>string</td></tr><tr><td>identityvalue</td><td>list< identity v1.8></td></tr><tr><td>defaultfris</td><td>list< fricurrencypairtype v1.8></td></tr><tr><td>loyaltypointsaccountfri</td><td>string</td></tr><tr><td>acceptedtc</td><td>acceptedtc v1.8</td></tr><tr><td>accountholderstatus</td><td>accountholderstatus v1.8</td></tr><tr><td>bankdomainname</td><td>string</td></tr><tr><td>homechargingregionname</td><td>string</td></tr><tr><td>accountholdertype</td><td>string</td></tr><tr><td>hasparent</td><td>boolean</td></tr><tr><td>languagecode</td><td>languagecode v1.8</td></tr><tr><td>child</td><td>list< accountholderbasicinfotype v1.3></td></tr></table>	Name	Type	firstname	string	surname	string	profilename	string	employeeid	string	identityvalue	list< identity v1.8>	defaultfris	list< fricurrencypairtype v1.8>	loyaltypointsaccountfri	string	acceptedtc	acceptedtc v1.8	accountholderstatus	accountholderstatus v1.8	bankdomainname	string	homechargingregionname	string	accountholdertype	string	hasparent	boolean	languagecode	languagecode v1.8	child	list< accountholderbasicinfotype v1.3>		
Name	Type																																		
firstname	string																																		
surname	string																																		
profilename	string																																		
employeeid	string																																		
identityvalue	list< identity v1.8>																																		
defaultfris	list< fricurrencypairtype v1.8>																																		
loyaltypointsaccountfri	string																																		
acceptedtc	acceptedtc v1.8																																		
accountholderstatus	accountholderstatus v1.8																																		
bankdomainname	string																																		
homechargingregionname	string																																		
accountholdertype	string																																		
hasparent	boolean																																		
languagecode	languagecode v1.8																																		
child	list< accountholderbasicinfotype v1.3>																																		
credentials	list< credentialtypedetails v1.1> Contains information about a credential (type and status). <table><tr><th>Name</th><th>Type</th></tr><tr><td>credentialtype</td><td>string</td></tr><tr><td>credentialstatus</td><td>credentialstatustype v1.1</td></tr></table> Credentialstatustype v1.1 <table><tr><th>Value</th><th>Description</th></tr><tr><td>ACTIVE</td><td>Credential is active.</td></tr><tr><td>ACTIVE_MUST_BE_CHANGED</td><td>Credential is active but needs to be changed.</td></tr><tr><td>BLOCKED</td><td>Credential is blocked.</td></tr><tr><td>INACTIVE</td><td>Credential is inactive.</td></tr><tr><td>SUSPENDED</td><td>Credential is temporary suspended.</td></tr></table>	Name	Type	credentialtype	string	credentialstatus	credentialstatustype v1.1	Value	Description	ACTIVE	Credential is active.	ACTIVE_MUST_BE_CHANGED	Credential is active but needs to be changed.	BLOCKED	Credential is blocked.	INACTIVE	Credential is inactive.	SUSPENDED	Credential is temporary suspended.	Mandatory	Contains information about a credential (type and status). Parameter can not be NULL. Validated with NotNullElements .														
Name	Type																																		
credentialtype	string																																		
credentialstatus	credentialstatustype v1.1																																		
Value	Description																																		
ACTIVE	Credential is active.																																		
ACTIVE_MUST_BE_CHANGED	Credential is active but needs to be changed.																																		
BLOCKED	Credential is blocked.																																		
INACTIVE	Credential is inactive.																																		
SUSPENDED	Credential is temporary suspended.																																		
offnet	boolean	Mandatory	True indicates that the party is an external user and not registered in the Mobile Money Manager. Parameter can not be NULL.																																

3.1.7 getsettlementdepositdetails v2.0

Retrieves the deposit details of a settlement.

Classification: **Undefined.**

URI: /{baseContext}/getsettlementdepositdetails

Namespace: http://www.ericsson.com/em/emm/settlement/v2_0

Parameters

getsettlementdepositdetailsrequest:

getsettlementdepositdetailsrequest			
Name	Type	Optionality	Description
bankcode	string	Optional	Custody account bank code. This along with custody account number may be used to identify



			the custody account. Validated with IsBankCode .										
accountnumber	string	Mandatory	Custody account number. This along with custody bank code will be used to identify the custody account. Parameter can not be NULL. Validated with IsBankAccount .										
numofdeposits	integer	Optional	The number of deposits to return in the response										
indexoffset	integer	Optional	Start retrieving deposits from this index.										
starttime	datetime v1.0	Optional	Select deposits from this time and onwards. Validated with IsDateTimeValue .										
endtime	datetime v1.0	Optional	Select deposits up until this time. Validated with IsDateTimeValue .										
filtertype	filtertype v1.0 Determines which field to filter on. <table><tr><th>Value</th><th>Description</th></tr><tr><td>BANKSTATUS</td><td>Filter on Bank instruction status</td></tr><tr><td>BANK_TRANSACTION_ID</td><td>Filter on Bank Transaction Id</td></tr><tr><td>SETTLEMENT_FILENAME</td><td>Filter on Settlement Filename</td></tr><tr><td>STATUS</td><td>Filter on instruction status</td></tr></table>	Value	Description	BANKSTATUS	Filter on Bank instruction status	BANK_TRANSACTION_ID	Filter on Bank Transaction Id	SETTLEMENT_FILENAME	Filter on Settlement Filename	STATUS	Filter on instruction status	Optional	String to filter results based on, either bank transaction id or settlement id.
Value	Description												
BANKSTATUS	Filter on Bank instruction status												
BANK_TRANSACTION_ID	Filter on Bank Transaction Id												
SETTLEMENT_FILENAME	Filter on Settlement Filename												
STATUS	Filter on instruction status												
filtervalue	string	Optional	String to filter results based on, either bank transaction id or settlement id. Validated with IsRestrictedString .										

getsettlementdepositdetailsresponse:

getsettlementdepositdetailsresponse																											
Name	Type	Optionality	Description																								
settlementdepositdetails	list< settlementdepositdetails v2.0> Contains information about a payment instruction response. <table><tr><th>Name</th><th>Type</th></tr><tr><td>id</td><td>long</td></tr><tr><td>settlementid</td><td>long</td></tr><tr><td>paymentinstructiondetails</td><td>paymentinstructiondetails v1.0</td></tr><tr><td>transactionid</td><td>string</td></tr><tr><td>result</td><td>paymentinstructionresult v2.0</td></tr><tr><td>message</td><td>string</td></tr><tr><td>failureReason</td><td>string</td></tr><tr><td>dateTime</td><td>dateTime</td></tr><tr><td>dateTime</td><td>dateTime</td></tr><tr><td>dateTime</td><td>dateTime</td></tr><tr><td>status</td><td>paymentinstructionresponsestatus v1.0</td></tr></table> Paymentinstructiondetails v1.0	Name	Type	id	long	settlementid	long	paymentinstructiondetails	paymentinstructiondetails v1.0	transactionid	string	result	paymentinstructionresult v2.0	message	string	failureReason	string	dateTime	dateTime	dateTime	dateTime	dateTime	dateTime	status	paymentinstructionresponsestatus v1.0	Optional	Contains a list of settlements. Contains information about a settlement deposit. Validated with NotNullElements .
Name	Type																										
id	long																										
settlementid	long																										
paymentinstructiondetails	paymentinstructiondetails v1.0																										
transactionid	string																										
result	paymentinstructionresult v2.0																										
message	string																										
failureReason	string																										
dateTime	dateTime																										
dateTime	dateTime																										
dateTime	dateTime																										
status	paymentinstructionresponsestatus v1.0																										



Name	Type
financialtransaction	long
identity	string
paymentinstructionid	long
externaltransactionid	string
calleridentifier	string
custodyaccountclearingnumber	string
custodyaccountnumber	string
dateTime	dateTime
amount	moneydetailstype v1.0
status	paymentinstructionstatus v1.0
accountholderclearingnumber	string
accountholderaccountnumber	string
fromfri	string
dateTime	dateTime

Paymentinstructionstatus v1.0

Value	Description
INACTIVE	The payment instruction is inactive.
PENDING	The payment instruction is pending.
PROCESSED	The payment instruction has been processed.
SENT_TO_BANK	The payment instruction is sent to bank.